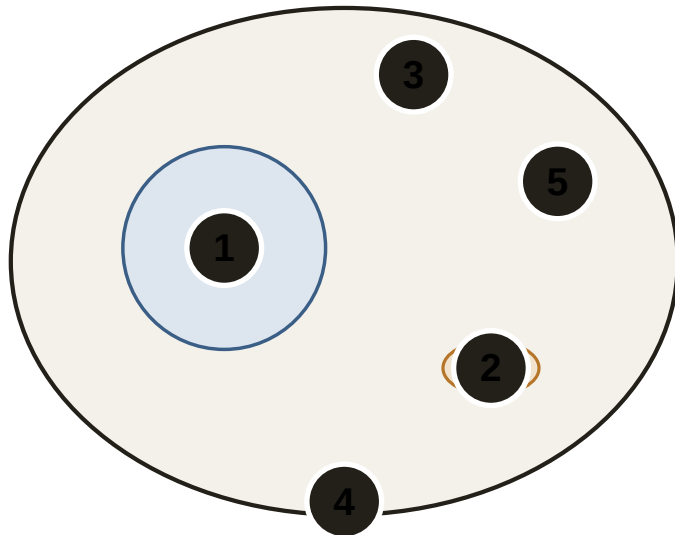


Animal & Plant Cells

Quick reference sheet · Biology · keep handy for revision

1. Animal cells

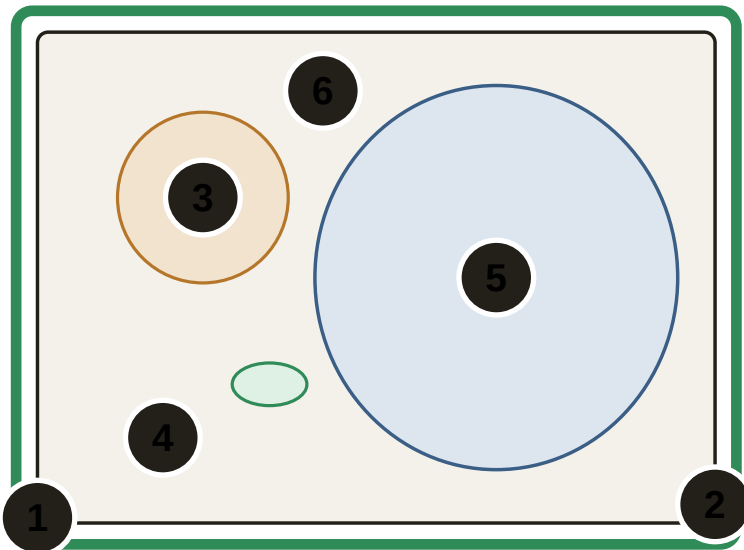
Most animal cells have the same five basic parts.



#	Structure	Function
1	Nucleus	Contains the genetic material (DNA), controls the cell's activities
2	Mitochondria	Site of respiration — releases energy the cell needs
3	Ribosomes	Where proteins are made
4	Cell membrane	Controls what substances enter and leave the cell
5	Cytoplasm	Jelly-like substance where many chemical reactions happen

2. Plant cells

Plant cells have everything an animal cell has, **plus** three extra structures.



#	Structure	Function
1	Cell wall	Made of cellulose ; strengthens and supports the cell
2	Cell membrane	Controls what substances enter and leave the cell
3	Nucleus	Contains the genetic material (DNA)
4	Chloroplast	Site of photosynthesis (contains chlorophyll)
5	Permanent vacuole	Filled with cell sap; helps keep the cell rigid
6	Cytoplasm	Jelly-like substance where many chemical reactions happen

3. Animal vs plant: what's the difference?

Structure	Animal	Plant
Nucleus, cytoplasm, cell membrane	✓	✓
Mitochondria, ribosomes	✓	✓
Cell wall (cellulose)	—	✓
Chloroplasts	—	✓
Permanent vacuole	—	✓

Plants can't move to escape danger or find support, so the **cell wall** gives each cell a rigid shape — millions of rigid cells stacked together let a plant stand upright. Only cells that photosynthesise (e.g. leaf cells) need **chloroplasts**; root cells, for example, have none.

4. A quick look at bacterial cells

Bacterial cells are much simpler and smaller than animal or plant cells. Their genetic material is not held in a nucleus — it is a single loop of DNA floating in the cytoplasm, plus small extra rings of DNA called **plasmids**.